Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	13
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 13)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2753.58	3100.5	1277.32	px^2
major_axis_length_px	lacuna shape	113.949	125.027	36.3344	px
minor_axis_length_px	lacuna shape	33.5852	33.3328	8.5152	px
aspect_ratio	lacuna shape	3.5038	3.3377	1.2462	unitless
eccentricity	lacuna shape	0.9377	0.9529	0.0506	unitless
solidity	lacuna shape	0.7478	0.7578	0.1342	unitless
roots_count	headline	17.3077	16	6.8846	count
ring_length_r30_px	headline	562.692	519	180.845	px
ring_length_r60_px	reported	1210.85	1176	368.189	px
owned_length_px	ownership-dependent	2614.05	2431.1	1219.35	px
edge_count	ownership-dependent	119.077	120	59.0134	count
mean_edge_length_px	ownership-dependent	22.3219	22.2102	2.5259	px
perimeter_px	normalised	322.901	322.043	98.3391	px
ring_area_r30_px2	normalised	11407.9	11663	2512.24	px^2
ring_area_r60_px2	normalised	27119.4	28286	5025.06	px^2
in_frame_fraction_r30	normalised	1	1	0.0001	unitless
in_frame_fraction_r60	normalised	0.9836	1	0.0403	unitless
ring_density_r30	normalised	0.0487808	0.0488869	0.00691877	px^-1
ring_density_r60	normalised	0.0439973	0.0424606	0.00657665	px^-1
roots_per_100px_perimeter	normalised	5.4494	5.5065	1.4975	per 100 px
ring_attached_length_r30_px	network v2	527.846	468	177.424	px
ring_attached_length_r60_px	network v2	1058.46	988	393.861	px
ring_length_w_r30_px	network v2	641.69	602.316	209.259	px
ring_length_w_r60_px	network v2	1377.16	1357.56	423.384	px
sholl_crossings_r10	network v2	12.4615	11	4.4275	count
sholl_crossings_r20	network v2	15	13	5.1478	count
sholl_crossings_r30	network v2	15.2308	15	4.7109	count
loop_count	loops, report only	9.8462	8	7.7443	count
loop_edge_share	loops, report only	0.2717	0.2895	0.1127	share
ring_width_mean_r30_px	width	7.4967	7.5581	0.3281	px

Per field
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measure	value	units
analysed_area_px2	1.00706e+06	px^2
total_skeleton_length_px	41354	px
canalicular_length_density_per_px	0.041064	px^-1
junction_count	967	count
junction_density_per_px2	0.00096022	px^-2
lacuna_count	17	count

lacunae_per_px2	1.688e-05	px^-2
skeleton_component_count	219	count
mean_component_length_px	188.831	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0465241	px^-1
field_density_without_flagged_per_px	0.0410058	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	344	px
unattached_component_count	18	count
canalicular_length_density_attached_per_px	0.041064	px^-1
field_length_density_with_unattached_per_px	0.0414056	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	167	count
loop_edge_share	0.3335	share
edge_count	2066	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2239	px
width_median_px	8.2462	px
width_p10_px	5.6569	px
width_p90_px	10.7703	px
width_pixels_used	40019	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	1001	62.1533	28.1242	2.21	0.8918	0.5533	-0.1968	55.7832	107.949	False
3	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
4	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
5	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
6	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
7	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
8	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
9	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
10	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
11	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
12	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
13	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
14	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
15	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True
16								310.754	226.539	False
17								933.644	3.9103	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	7	218	587	843.269	34	24.802	7.8744
2	14	387	707	1470.98	54	27.2404	7.568
3	11	329	812	1795.26	74	24.2603	6.9416
4	10	519	1176	3709.11	167	22.2102	7.5297
5	30	842	1719	4824.39	205	23.5336	7.5581
6	16	445	901	1248.21	57	21.8983	7.4017
7	16	480	903	1053.81	51	20.6629	7.6632
8	7	420	1056	2124.43	89	23.87	8.2449
9	17	525	1324	3737.76	152	24.5906	7.655
10	28	922	1954	4239.94	236	17.9659	7.1983
11	22	644	1381	2431.1	120	20.2591	7.0473
12	20	650	1360	2447.77	125	19.5821	7.6042
13	12	433	976	3230.66	136	23.7549	7.3677
14	9	405	936	3381.85	140	24.1561	7.6782
15	6	240	595	278.936	13	21.4566	5.8744
16	22	719	1472	1669.18	82	20.3558	7.6774
17	3	112	250	494.132	25	19.7653	7.4358

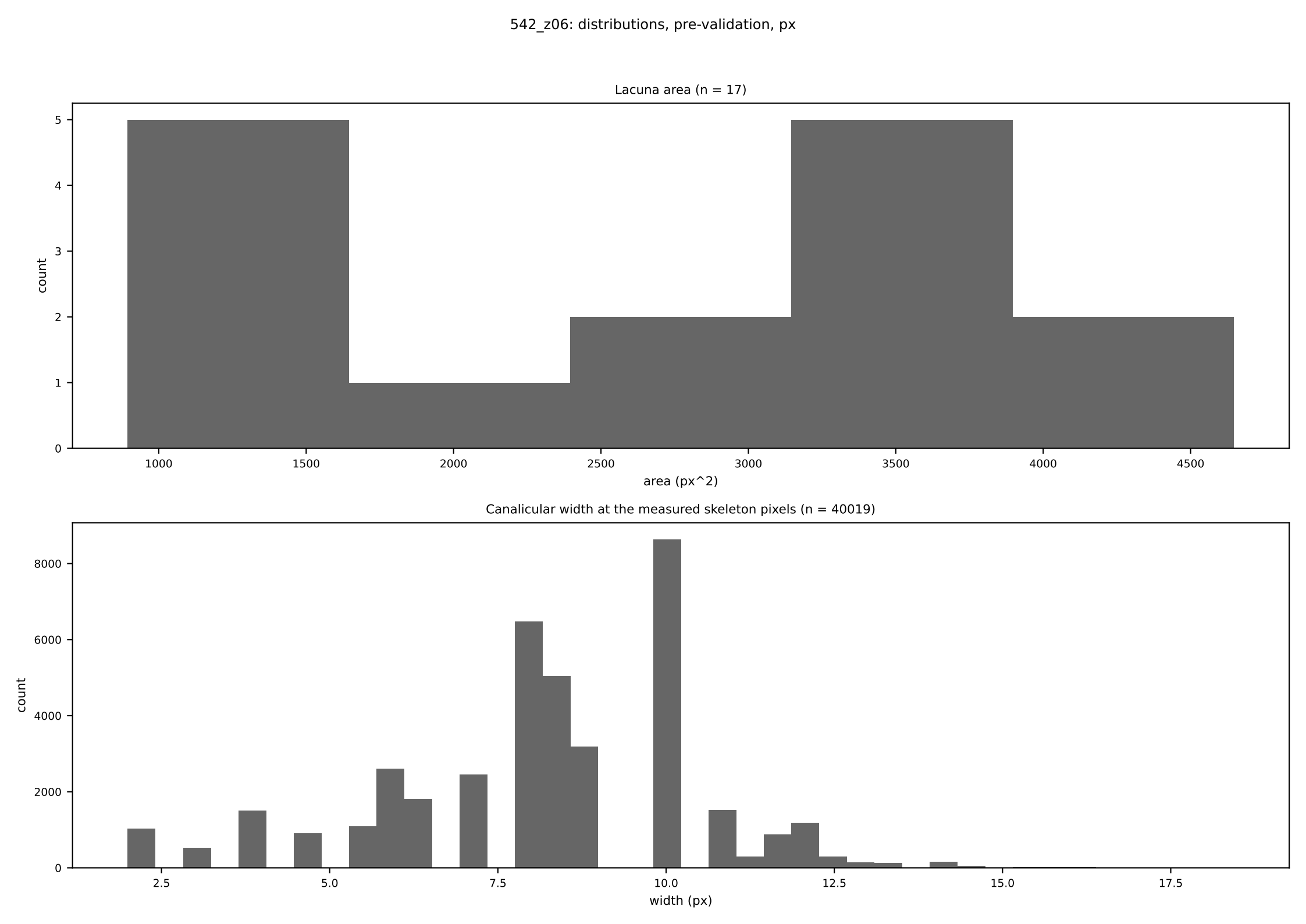
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0388107	0.0462423	3.2523
2	225.309	8379	19327	0.9998	0.884	0.0461869	0.036581	6.2137
3	151.782	6809	18568	1	1	0.0483184	0.0437312	7.2472
4	322.043	11663	28685	1	1	0.0444997	0.040997	3.1052
5	438.517	13901	31616	1	1	0.0605712	0.0543712	6.8412
6	211.924	8845	21628	1	1	0.0503109	0.041659	7.5499
7	359.191	12383	26961	1	0.9025	0.0387628	0.0334928	4.4545
8	302.451	11295	27170	1	1	0.0371846	0.0388664	2.3144
9	309.522	11337	28286	1	1	0.0463085	0.0468076	5.4923
10	468.274	15363	34096	1	1	0.0600143	0.0573088	5.9794
11	422.978	13165	30001	1	1	0.0489176	0.0460318	5.2012
12	379.848	13296	32108	1	1	0.0488869	0.042357	5.2653
13	217.924	8830	22986	1	1	0.0490374	0.0424606	5.5065
14	332.894	10505	23806	0.8602	0.7919	0.0385531	0.0393178	2.7036
15	205.776	6766	16357	0.7841	0.714	0.0354715	0.0363759	2.9158
16	387.948	13037	31120	1	1	0.0551507	0.0473008	5.6709
17	87.0061	2915	8653	0.5433	0.5282	0.038422	0.0288917	3.448

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	580	251.48	670.115	6	5	5
2	327	652	444.647	803.896	9	7	10
3	325	675	372.492	918.938	6	7	10
4	468	988	607.428	1357.56	11	17	15
5	839	1624	970.749	1977.11	23	20	22
6	406	778	502.404	1019.81	9	13	13
7	436	711	539.375	1011.08	11	13	10
8	367	650	473.647	1182.69	12	13	13
9	525	1315	602.316	1518.33	11	13	16
10	853	1877	1058.09	2213.7	17	23	25
11	602	1210	727.985	1563.67	16	17	15
12	617	1262	736.227	1539.43	15	23	19
13	433	846	487.433	1106.16	9	12	12
14	389	799	459.99	1056.36	13	11	16
15	224	259	273.35	667.428	7	7	8

16	664	1172	819.169	1690.7	13	17	18
17	112	241	127.74	281.723	3	3	5


542_z06: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 40019)
count
width (px)